Binomial Probability Distribution

$$P(x=k) = C_k^n p^k q^{n-k} = \frac{n!}{k!(n-k)!} p^k q^{n-k}$$

$$\text{Mean} : \mu = np$$

$$\text{Variance} : \sigma^2 = npq$$

$$\text{Standard deviation} : \sigma = \sqrt{npq}$$

Hypergeometric Probability Distribution

$$P(x=k) = \frac{C_k^M C_{n-k}^{N-M}}{C_n^N}$$

M → successes, ***N-M*** → failures,
n → size of the random sample space

$$\text{Mean} : \mu = n\left(\frac{M}{N}\right)$$

$$\text{Variance} : \sigma^2 = n\left(\frac{M}{N}\right)\left(\frac{N-M}{N}\right)\left(\frac{N-n}{N-1}\right)$$

Poisson Probability Distribution

$$P(x=k) = \frac{\mu^k e^{-\mu}}{k!}$$

Mean: $E(x) = \mu$

$$\text{Variance} : \sigma^2 = \mu$$

Standard deviation: $\sigma = \sqrt{\mu}$

Variance of a Sample

$$s^2 = \frac{\sum(x_i - \bar{x})^2}{n-1} = \frac{\sum x_i^{\,2} - \dfrac{(\sum x_i)^2}{n}}{n-1}$$

Variance of Population: $\sigma^2 = \dfrac{\sum(x_i - \mu)^2}{N}$

Correlation coefficient, $r = \dfrac{s_{xy}}{s_x s_y}$

Covariance, $s_{xy} = \dfrac{\sum(x_i - \bar{x})(y_i - \bar{y})}{n-1}$, or

$$s_{xy} = \frac{\sum x_i y_i - \dfrac{(\sum x_i)(\sum y_i)}{n}}{n-1}$$

Regression line, $y = a + bx$

$$b = r\frac{s_y}{s_x} \quad , \quad a = \bar{y} - b\bar{x}$$

Standardizing the value of x :

$z = \dfrac{\boldsymbol{x} - \boldsymbol{\mu}}{\boldsymbol{\sigma}}$, or in a sample, $z = \dfrac{\boldsymbol{x} - \bar{\boldsymbol{x}}}{\boldsymbol{s}}$